Logic Design and Switching Theory

Lecture – 12

Subtitle: Shift Registers

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Previous

Counters

- Asynchronous
- synchronous

Today's Topic

Shift Registers

Shift Register

A shift register is a type of digital memory circuit used in electronics and computer systems to store and shift data.

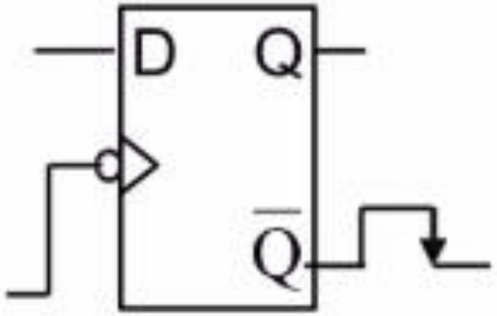
It consists of a series of flip-flops connected in a chain, where the output of one flip-flop becomes the input of the next.

Shift registers are commonly used for data storage, transfer, and manipulation.

Shift Register

- Shift registers are constructed using several flip- flop, connected in such a way to STORE and TRANSFER/ Shift digital data.

- Basically, D flip-flop is used. The input data (either '0' or '1') is applied to the D terminal and the data will be stored at Q during positive/negative-edge transition of the clock pulse.



Terminology

Serial Input (SI): The input line through which data is entered into the shift register one bit at a time. E.g. You input a stream like 1 → 0 → 1 → 1 into a 4-bit register, one bit per clock.

Serial Output: The register outputs data one bit at a time, typically from one end of the register (e.g., LSB or MSB first).

Use case: Sending data over a single wire or line (like in SPI communication).

Parallel Input: All bits are loaded into the register simultaneously (at once).

Example: You load 1011 into a 4-bit register in a single operation.

Parallel Output: The output lines through which the stored data in the shift register is accessed simultaneously, with each output line corresponding to a flip-flop in the register.

Type	Description
Serial-In Serial-Out (SISO)	Data enters and exits one bit at a time (used for data buffering).
Serial-In Parallel-Out (SIPO)	Data enters one bit at a time and can be read all at once.
Parallel-In Serial-Out (PISO)	Multiple bits loaded at once, then shifted out one by one.
Parallel-In Parallel-Out (PIPO)	All bits loaded and read simultaneously.
Bidirectional Shift Register	Can shift data left or right, based on a control signal.
Universal Shift Register	Supports all operations: serial, parallel, bidirectional shifts.

Type	Input	Output	Use Case
SISO	Serial	Serial	Delay line, serial data transfer
SIPO	Serial	Parallel	Serial to parallel conversion
PISO	Parallel	Serial	Parallel to serial conversion
PIPO	Parallel	Parallel	Fast data storage and retrieval

Mode	Clocks need for n-bit shift register		
	Loading	Reading	Total
SISO	N	N-1	2n-1
SIPO	N	0	N
PISO	1	N-1	N
PIPO	1	0	1

Applications

- Data conversion (serial ↔ parallel)
- Time delays in digital circuits
- Signal processing
- LED matrix control
- Finite state machines (FSM)

Serial-In Serial-Out (SISO)

Function:

- Data is loaded bit-by-bit in serially (one at a time).
- Each clock pulse shifts the data to the next flip-flop.
- Data is read out serially, in the same order (first in, first out).

Example:

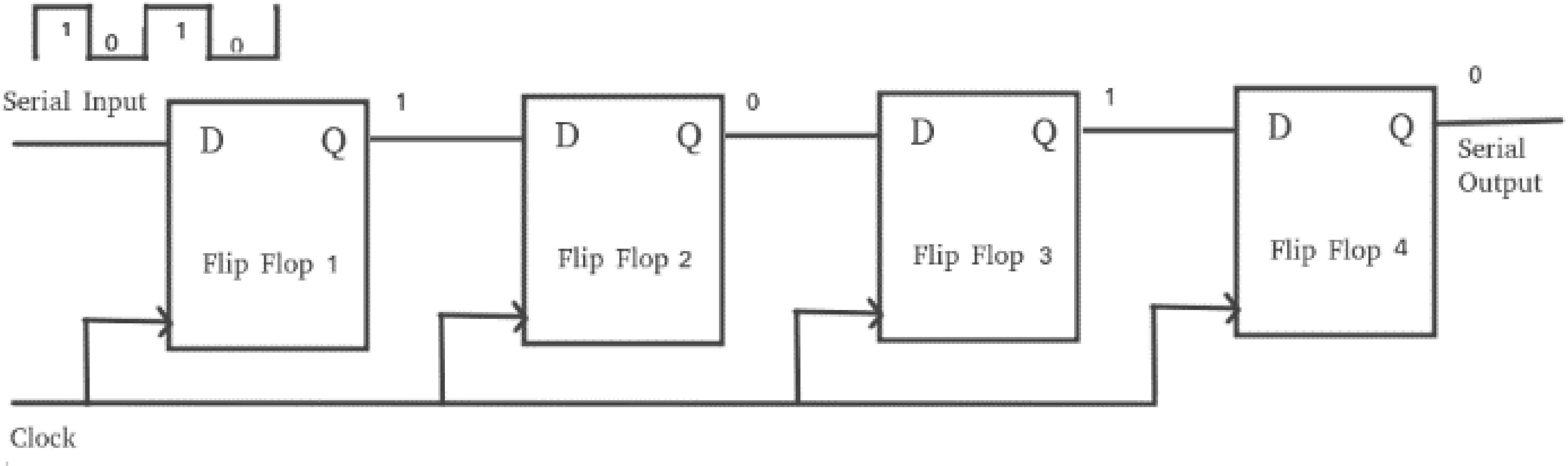
- Delay lines in digital signal processing.

Serial-In Serial-Out (SISO)

A Serial-In Serial-Out shift register is a sequential logic circuit that allows data to be shifted in and out one bit at a time in a serial manner.

It consists of a cascade of flip-flops connected in series, forming a chain. The input data is applied to the first flip-flop in the chain, and as the clock pulses, the data propagates through the flip-flops, ultimately appearing at the output.

Serial-In Serial-Out (SISO)



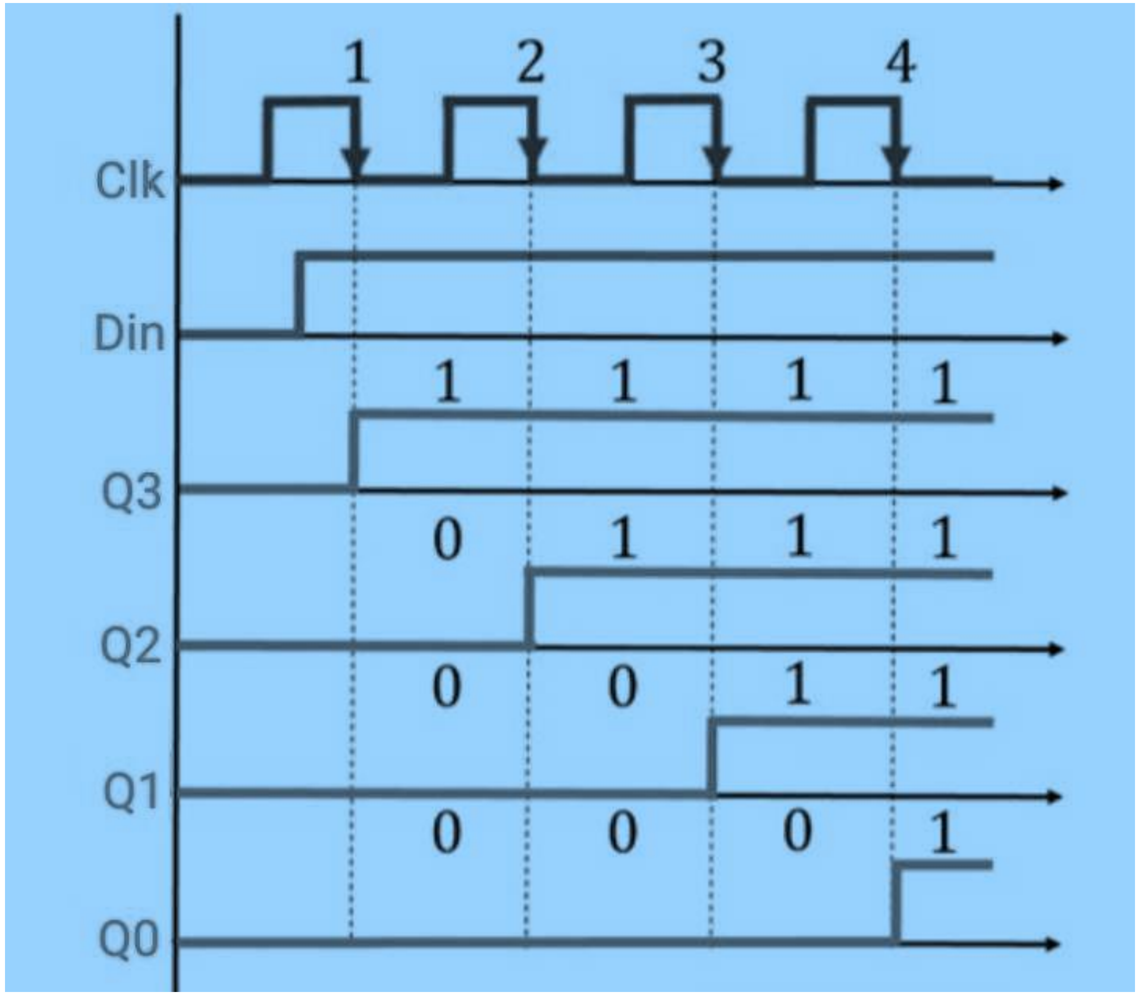
4 Bit SISO Register

Serial-In Serial-Out (SISO)

The operating principle of SISO involves feeding the intended signal into the system where it gets processed in order to produce an output. Therefore, input signal refers to what goes into the system while output signal describes the result or response produced by the system depending on that particular input.

Signals at both the input and the output can be mathematically modeled by functions or equations in a SISO system. A system can be represented through a transfer function, describing how the system will modify the input signal to produce the output signal.

Serial-In Serial-Out (SISO)



CLK	Q3	Q2	Q1	Q0
Initial (Reset)	0	0	0	0
After 1st clock pulse	1	0	0	0
After 2nd clock pulse	1	1	0	0
After 3rd clock pulse	1	1	1	0
After 4th clock pulse	1	1	1	1

Applications of SISO Shift Registers

- **Data Storage and Retrieval**: SISO shift registers are often used to store and retrieve data in applications where serial transmission is more efficient or feasible. For instance, in communication systems, data is often transmitted serially, and a shift register allows for temporary storage before further processing.
- **Serial-to-Parallel Conversion**: By using a SISO shift register in combination with additional circuitry, serial data can be converted into parallel form.
- **Delay and Time-Sequence Generation**: The cascaded nature of shift registers enables the generation of delayed versions of a signal or the generation of complex time sequences. By tapping into different stages of the shift register, various delayed versions of the input signal can be obtained.
- **Data Encryption and Decryption**: Shift registers can be employed in cryptographic applications for data encryption & decryption.
- **Frequency Division and Counting**: Serial-In Serial-Out shift registers are used in frequency division and counting applications. By connecting the output of a flip-flop back to the input, the shift register can function as a binary counter, dividing the input frequency by a factor of two with each clock pulse.

Disadvantages of SISO Register

- **Slower Data Transmission :** Due to processing data series ways, SISO shift registers may have lower data transfer.
- **Limited Capability for Parallel Processing :** The SISO Shift Registers process one bit at a time, thus restrict the ability to parallel processing. They are not suitable for applications where parallel processing is of the essence.
- **Complexity for Performing Multiple Operations :** In case an application requires both sequential and parallel logic operations simultaneously on multi-bit inputs, this need may become impractical since several SISO will be required rather than a single one thus increasing complexity and making it less efficient than using a PIS register.
- **Data Synchronization Difficulties :** As far as synchronization with respect to the synchronization input and output is concerned, it becomes difficult due to sequential nature of SISO shift registers in systems where such aspects are crucial.
- **High propagation delays :** When using a SISO register, we will often see higher bureaucratic losses should we try transmitting data through it too fast because it just cannot do such things like an analog circuit would have been able to perform those actions.

Serial-In Parallel-Out (SIPO)

Function:

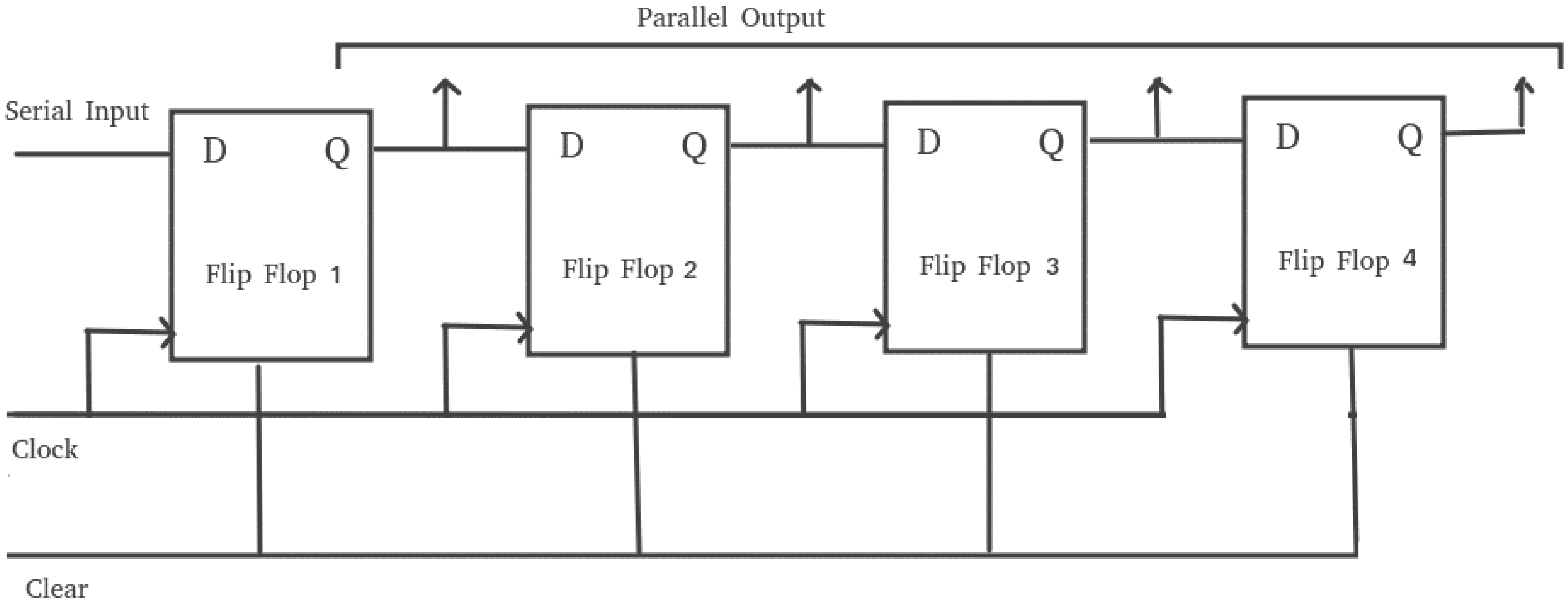
- Data is input serially (bit-by-bit).
- After a few clock pulses, all bits are available at once on parallel output lines.

Example:

- Sending serial data from a microcontroller to control multiple LEDs.

Serial-In Parallel-Out (SIPO)

The SIPO shift register enables serial data input and parallel data output, making it useful for various applications, such as data buffering, data acquisition, and control systems.



Operation Modes

- **Shift Mode**: In shift mode, the input data is serially shifted through the flip-flops. The data bit at the serial input enters the first flip-flop, while the existing data bits are shifted to subsequent flip-flops. This mode is useful when serial data needs to be processed sequentially.
- **Parallel Load Mode**: In parallel load mode, the shift register holds the data in a fixed state, allowing parallel loading of data. The data bits are directly applied to the parallel inputs of each flip-flop. When the clock signal is activated, the data is latched into the respective flip-flops simultaneously. This mode is beneficial when data needs to be loaded in parallel.

Applications of SIPO Shift Registers

Data Storage and Buffering: They allow serial data to be converted into parallel data, ensuring smooth communication between different components of a system.
Address Decoding: SIPO shift registers are used for address decoding in memory devices and microprocessor. They allow efficient selection of memory locations and enable data retrieval or storage based on the specific address.
Control Systems: In control systems, shift registers are utilized to store and shift control signals, enabling sequential operations and timing synchronization. They are often used for generating timing sequences, control signals, and state machine implementations.

Parallel-In Serial-Out (PISO)

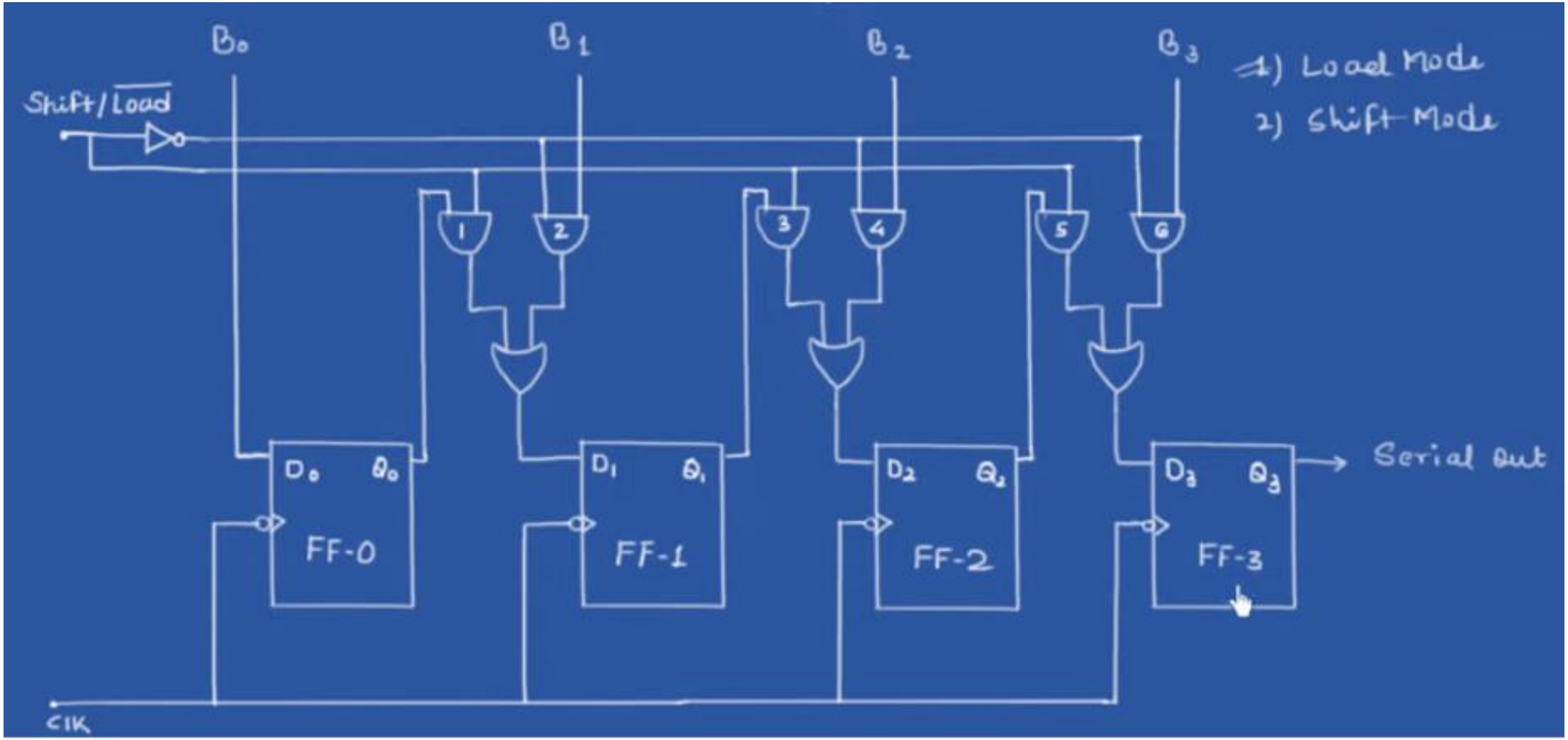
Function:

- Multiple bits are loaded in parallel into flip-flops at once.
- The bits are then shifted out one by one (serially).

Example:

- Keyboard scanning or sending sensor data to a serial port.

Parallel-In Serial-Out (PISO)



Parallel-In Parallel-Out (PIPO)

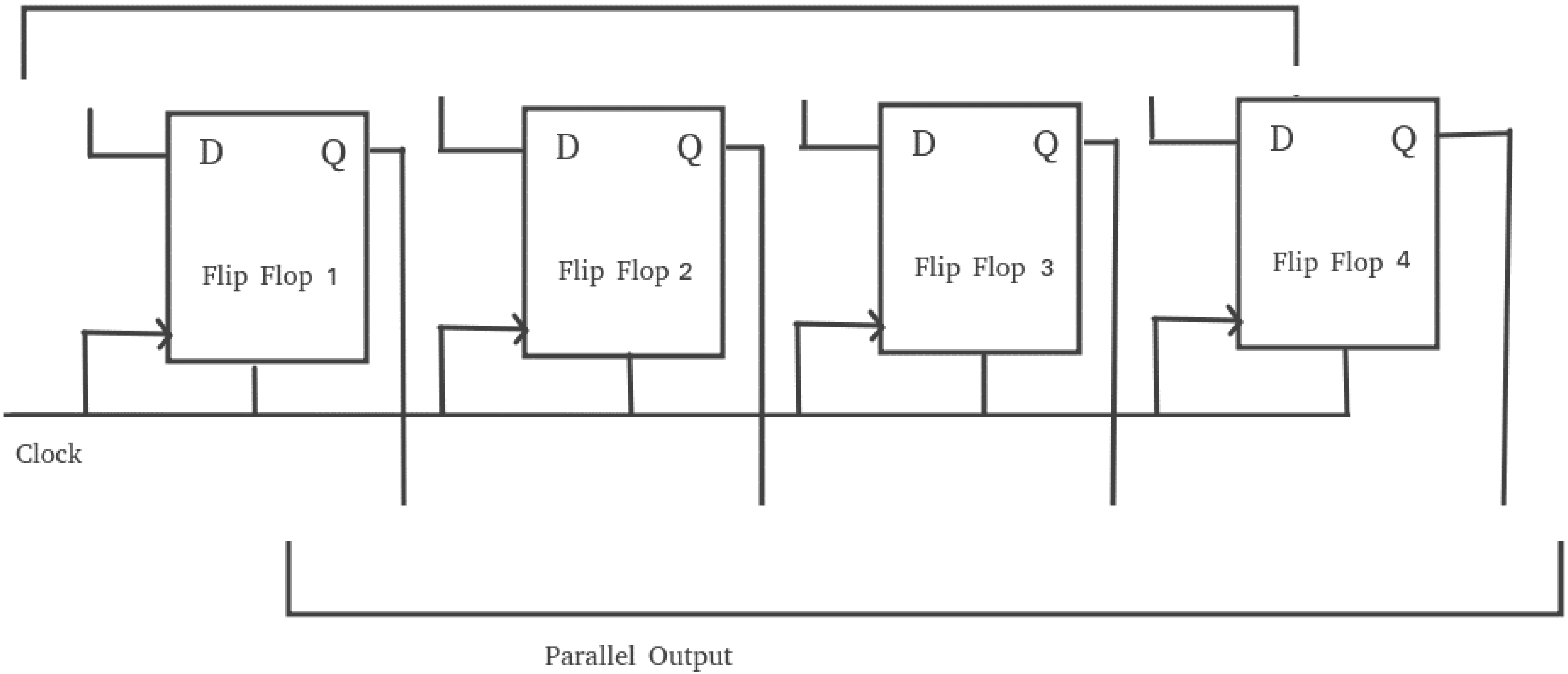
Function:

- All bits are loaded and made available simultaneously on the outputs.
- Not actually used for shifting—just parallel loading and reading.

Example:

- Temporary data storage or bus interfacing.

Parallel-In Parallel-Out (PIPO)



Bidirectional shift register

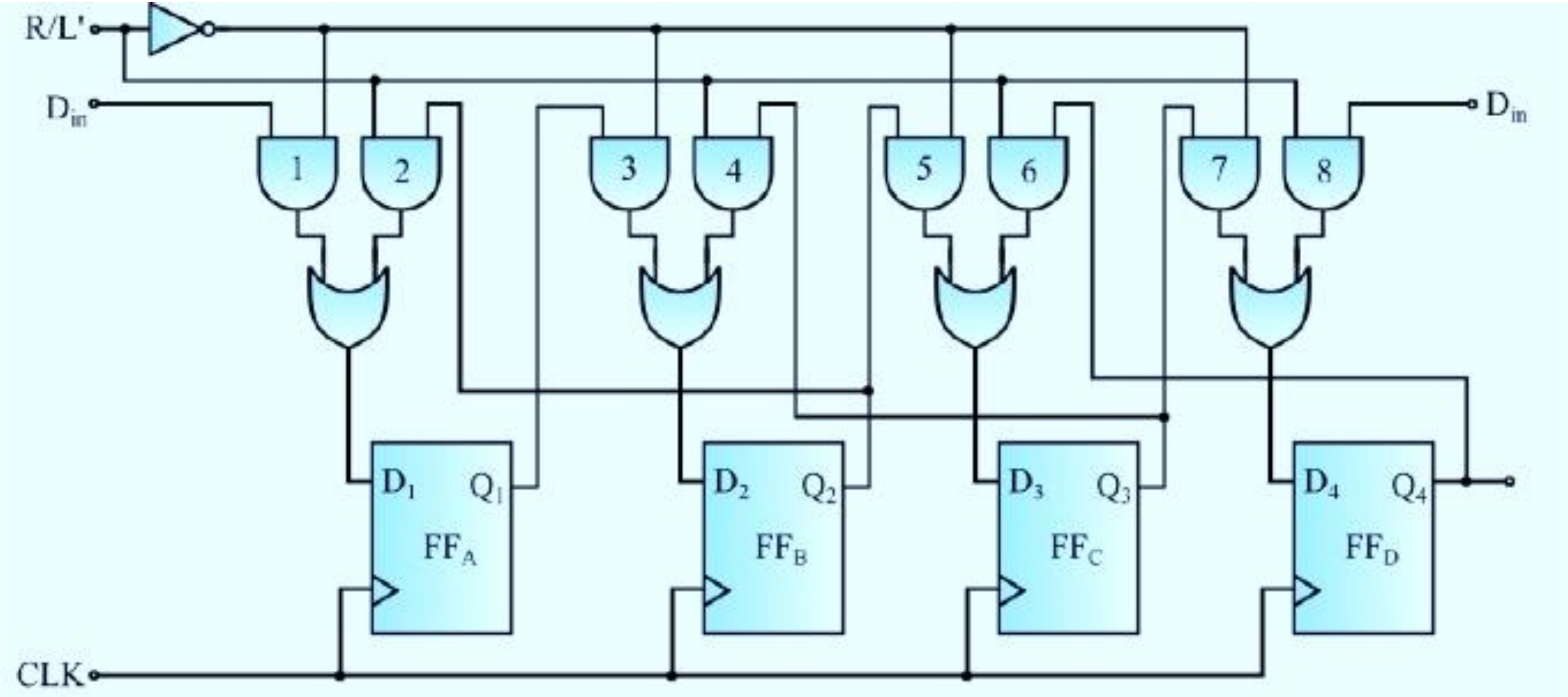
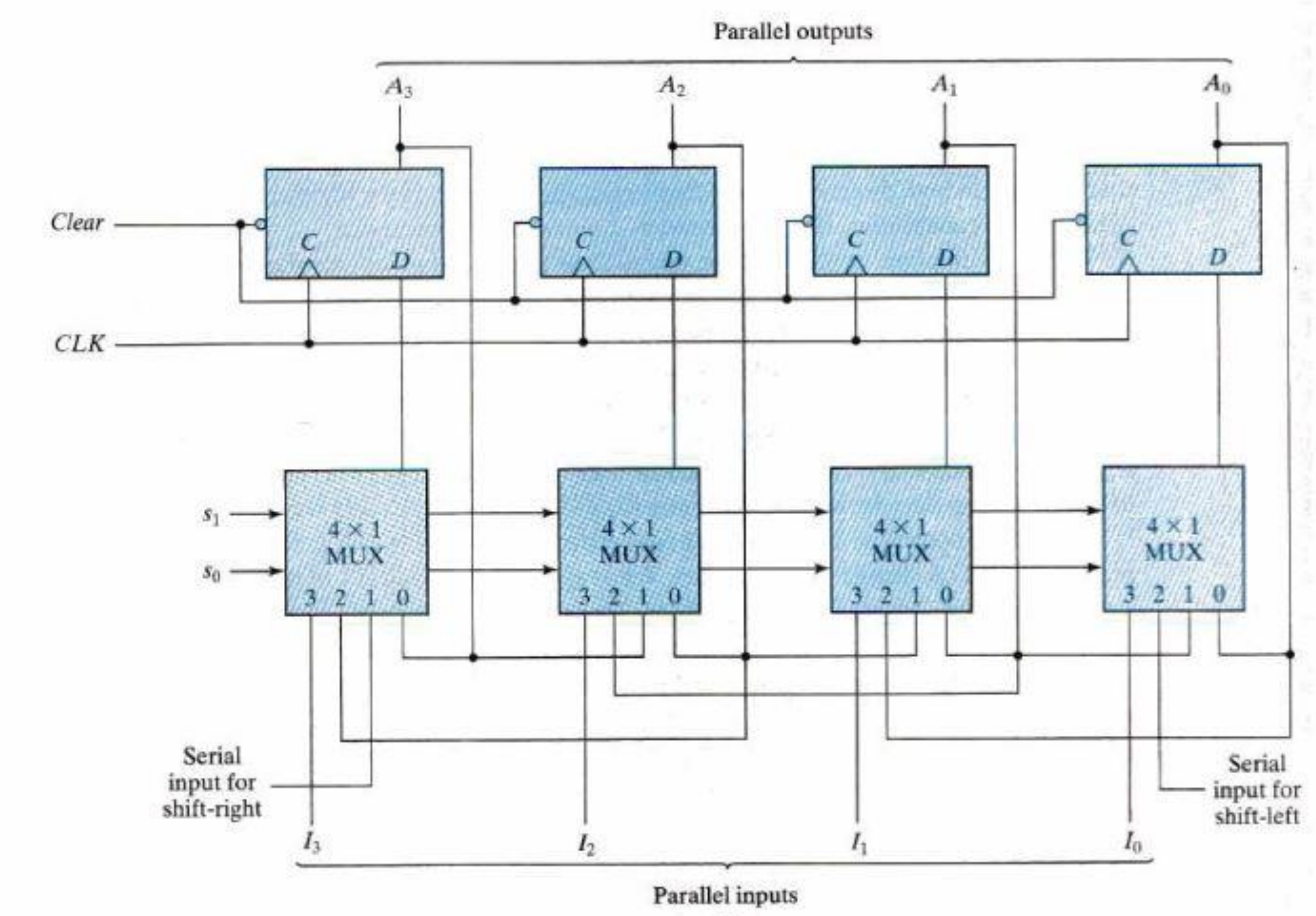


Figure - Bidirectional Shift Register

Bidirectional Universal shift register



Thank you!